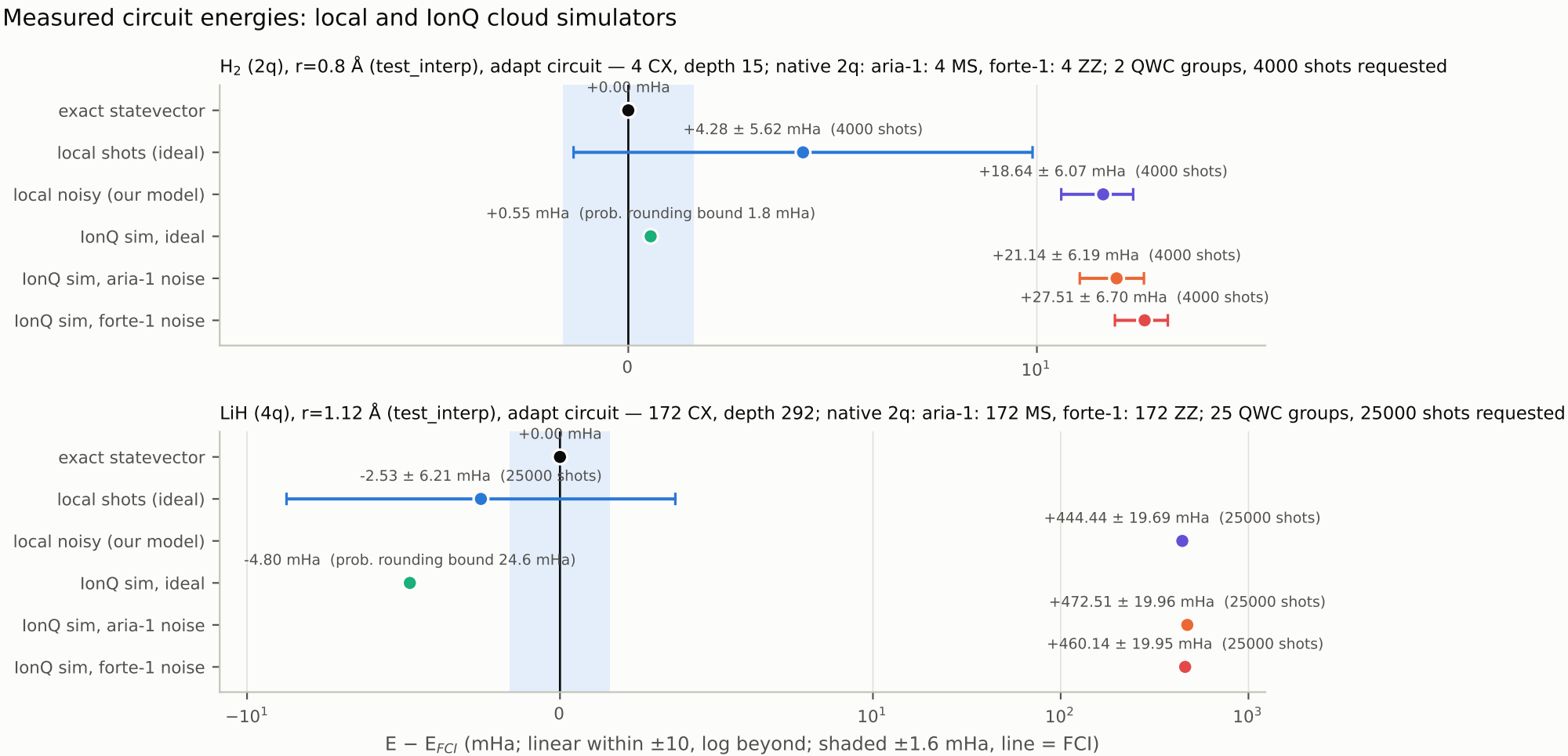
Measured circuit energies: local and IonQ cloud simulators
H2 (2q), r=0.8 Å (test_interp), adapt circuit — 4 CX, depth 15; native 2q: aria-1: 4 MS, forte-1: 4 ZZ; 2 QWC groups, 4000 shots requested
exact statevector
local shots (ideal)
local noisy (our model)
IonQ sim, ideal
IonQ sim, aria-1 noise
IonQ sim, forte-1 noise
+0.00 mHa
+4.28 ± 5.62 mHa (4000 shots)
+18.64 ± 6.07 mHa (4000 shots)
+0.55 mHa (prob. rounding bound 1.8 mHa)
+21.14 ± 6.19 mHa (4000 shots)
+27.51 ± 6.70 mHa (4000 shots)
0
10^1
LiH (4q), r=1.12 Å (test_interp), adapt circuit — 172 CX, depth 292; native 2q: aria-1: 172 MS, forte-1: 172 ZZ; 25 QWC groups, 25000 shots requested
exact statevector
local shots (ideal)
local noisy (our model)
IonQ sim, ideal
IonQ sim, aria-1 noise
IonQ sim, forte-1 noise
+0.00 mHa
-2.53 ± 6.21 mHa (25000 shots)
+444.44 ± 19.69 mHa (25000 shots)
-4.80 mHa (prob. rounding bound 24.6 mHa)
+472.51 ± 19.96 mHa (25000 shots)
+460.14 ± 19.95 mHa (25000 shots)
−10^1
0
10^1
10^2
10^3
E − E_FCI (mHa; linear within ±10, log beyond; shaded ±1.6 mHa, line = FCI)